

GUSSEPPE BRAVO

AI Research Engineer | PhD in AI

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 gussepebravo.com

 gussepebravo

 gussepe

 New York, USA · Open to relocation

"If I can't draw it, I can't get it"

SUMMARY

- AI Research Engineer with PhD (highest honors) specializing in ML systems that adapt to real-world change. Built first-of-its-kind AI agents for production model monitoring and improvement (CAMA, KC-Agent)
- 6+ years at Barcelona Supercomputing Center/Lenovo developing continual learning frameworks for autonomous vehicles. Currently building healthcare cognitive agents at Cotiviti and contributed to diabetic retinopathy research at Emory Medical Center
- Published in top-tier AI venues (AAMAS, ICPR, VISAPP). PhD research achieved 99% storage reduction while maintaining 96.9% accuracy in privacy-sensitive healthcare applications through autonomous prompt evolution
- Delivered measurable impact: \$200K research funding, \$100K integration grants, \$10M+ increased tax revenue, and real-world deployments across healthcare, autonomous driving, and enterprise analytics
- Framework-agnostic prototyping on Linux. Rapid iteration from concept to production. Expert at distilling complexity by removing noise and solving hard problems with minimal overhead

EXPERIENCE

Generative AI Research Engineer (Intern)
Cotiviti

 Jul 2024 – Present  New York, USA

- Deploying agentic RAG systems. I built a cost-efficient pipeline for parsing and retrieving complex documents (1.5k for 0.5\$)
- Building continual learning cognitive architecture that enables agents to learn from interaction.

AI Research Engineer
Barcelona Supercomputing Center & Lenovo

 Feb 2019 – December 2025  Spain/USA

- Architected cognitive frameworks for Language Agents that autonomously monitor ML model health, reducing degradation incidents and improving model reliability
- Built TADIL algorithm for domain-agnostic continual learning, enabling 6% performance improvement across diverse domains without requiring task labels
- Developed Scanflow, an MLOps platform for debugging ML workflows, some projects were built on top of it.
- Developed a novel drift detection algorithm using autoencoder for drift detection on images. Published in top-tier journal.
- Led cross-functional team, resulting in several publications in top-tier journals and conferences

AI Engineer Intern

CORE COMPETENCIES

Production AI Systems	
Python/Linux/MLOps	
Continual Learning	
Problem solving	

STACK SOFTWARE

Linux	Python	PyTorch	Numba
LangGraph	AutoGen	Langchain	
transformers	MLflow	Ray	Docker
Kubernetes	RAG	Spark	LLMs

IMPACT METRICS

	\$250K+ Research Funding Secured from Lenovo and Intel for AI research initiatives
	\$10M+ Revenue Impact Generated for Peruvian Tax Administration through AI fraud detection
	40+ GitHub repos For open-source ML/MLOps projects and libraries
	10+ Publications In top-tier AI journals and conferences (AAMAS, ICPR, CVPR)

EDUCATION

PhD in Artificial Intelligence
UPC BarcelonaTech

 2021 – 2025

Research focus: Human-on-the-loop Continual Learning: Data, Knowledge and Agents for Model Adaptation **Highest honors: Cum laude**

MSc in Artificial Intelligence
UPC BarcelonaTech

 2019 – 2021

Emory Global Diabetes Research Center

Jan 2024 – Jul 2024 USA

- Developed resource-efficient **cognitive architecture** for LLM-agents using finetuned small language models (Llama, Phi), **reducing API costs by 35%** while maintaining reasoning capabilities
- Built privacy-preserving **domain adaptation algorithm** for diabetic retinopathy detection, achieving **96.9% accuracy with 99% storage reduction** across diverse patient populations
- Published 3 papers on continual learning for medical imaging, including **Best Paper Nominee at VISAPP 2025**

AI Consultant

SUNAT-Inter-American Development Bank

Jan 2022 – Dec 2025 Peru

- Built the **first agent-based system** for customs to monitor ML models in production, enabling **real-time detection of customs fraud patterns**
- Built the first **"Improver" agent for automated ML model updates** that **reduced manual intervention by 65%** while maintaining model performance
- Created SmartDeploy, the first end-to-end MLOps system for SUNAT, enabling **continuous delivery of ML models** for tax evasion detection

Data Scientist

SUNAT (Tax Administration)

Nov 2017 – Nov 2018 Peru

- Engineered ensemble-based fraud classifier algorithm with **92% accuracy** for identifying unjustified patrimony
- Developed graph analysis methods to uncover complex tax evasion networks, **processing 1TB+ of financial data** monthly
- Built scalable analytical infrastructure processing millions of electronic invoices, directly contributing to **\$10M increase** in tax revenue

KEY PROJECTS & OPEN SOURCE

Cognitive Architecture for ML Monitoring

github.com/gussepe/cognitive_architecture_checker

- LLM-based agent framework that **autonomously monitors ML models** in production environments
- Developed novel **decision procedure algorithm** and **structured memories** for interpretable model diagnosis
- Featured in **AAMAS 2025 conference**; developed in collaboration with Lenovo AI Lab

Scanflow

github.com/gussepe/scanflow

- Production-grade MLOps** framework for ML workflow management, debugging, and reliability
- Many projects were inspired by this framework, like Scanflow-k8s, CAMA, KC-Agent.
- Published in top-tier journal "Expert Systems With Applications" (Impact Factor: 8.665)

Thesis: "Scanflow: A Learning-symbolic Framework for ML Workflow Debugging" **Published in Expert Systems With Applications (IF: 8.665)**

BSc in Computer Science, Magna Cum Laude

National University of Engineering

2011 – 2016

Top 5% of graduating class

AWARDS

 **PhD highest honors (2025)**
Cum laude for work on Continual Learning and Agents

 **Intel-Lenovo AI Innovators Award (2023)**
Selected for innovation in continual learning and cognitive architectures research

 **María Rostworowski I Researcher (2020)**
National recognition for excellence in AI research

SOME PUBLICATIONS

Feature Engineering for Agents
AAMAS 2025

Experience Replay and Zero-shot Clustering
VISAPP 2025

TADIL: Task-Agnostic Domain Incremental Learning
ICPR 2024

Scanflow: Multi-graph framework
Expert Systems With Applications

LANGUAGES

English ● ● ● ● ●

Spanish ● ● ● ● ●